

Residual-Aware Bayesian Clustering:

Complete Derivations from First Principles

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1 Probability Densities and Likelihood

1.1 The Gaussian density

A d -dimensional Gaussian (normal) distribution with mean $\mu \in \mathbb{R}^d$ and covariance $\Sigma \in \mathbb{R}^{d \times d}$ (symmetric positive definite) has density

$$\mathcal{N}(x; \mu, \Sigma) = \frac{1}{(2\pi)^{d/2} |\Sigma|^{1/2}} \exp\left(-\frac{1}{2}(x - \mu)^\top \Sigma^{-1}(x - \mu)\right). \quad (1)$$

- The normalizer $(2\pi)^{d/2} |\Sigma|^{1/2}$ ensures $\int_{\mathbb{R}^d} \mathcal{N}(x; \mu, \Sigma) dx = 1$.
- The exponent $-\frac{1}{2}(x - \mu)^\top \Sigma^{-1}(x - \mu)$ is a quadratic form measuring “distance” from μ in the metric defined by Σ^{-1} . Points far from μ (in the Σ -metric) get low density.
- The **precision matrix** Σ^{-1} controls the shape: large eigenvalues of Σ^{-1} (small eigenvalues of Σ) mean the distribution is narrow in that direction.

Intuition 1.1. The Gaussian is the maximum-entropy distribution for a given mean and covariance. If all you know about a random variable is its mean and spread, the Gaussian is the “least-assuming” density consistent with those constraints. This is why it appears everywhere as a default model for continuous noise.

1.2 Likelihood: turning a distribution into a score

Given a fixed observation $x_{\text{obs}} \in \mathbb{R}^d$, we can *evaluate* the density at that point:

$$\ell(\mu, \Sigma) = \mathcal{N}(x_{\text{obs}}; \mu, \Sigma). \quad (2)$$

This number ℓ is called the **likelihood** of the parameters (μ, Σ) given the data x_{obs} . Note:

- The observation x_{obs} is **fixed** (it’s the data we collected).
- The parameters (μ, Σ) are **variable** (we want to find the best ones).
- The likelihood is *not* a probability distribution over (μ, Σ) —it does not integrate to 1 over parameter space.

With P independent observations $x_1, \dots, x_P$, the joint likelihood is the product:

$$\ell(\mu, \Sigma) = \prod_{i=1}^P \mathcal{N}(x_i; \mu, \Sigma). \quad (3)$$

Taking the logarithm (which preserves the location of the maximum):

$$\log \ell(\mu, \Sigma) = \sum_{i=1}^P \log \mathcal{N}(x_i; \mu, \Sigma) = -\frac{Pd}{2} \log 2\pi - \frac{P}{2} \log |\Sigma| - \frac{1}{2} \sum_i (x_i - \mu)^\top \Sigma^{-1} (x_i - \mu). \quad (4)$$

Key idea. Maximum likelihood estimation (MLE): choose (μ, Σ) to maximize $\log \ell$. For a single Gaussian, the MLE is the sample mean $\hat{\mu} = \frac{1}{P} \sum_i x_i$ and sample covariance $\hat{\Sigma} = \frac{1}{P} \sum_i (x_i - \hat{\mu})(x_i - \hat{\mu})^\top$. This is tractable and has a unique closed-form solution.

1.3 Conditional distributions

A **conditional distribution** $p(y | x)$ is a density over y that depends on a given value of x . It satisfies $\int p(y | x) dy = 1$ for each fixed x . The dependence on x enters through the parameters of the density.

Example 1.1 (Linear regression). $y | x \sim \mathcal{N}(w^\top x, \sigma^2)$. The mean $w^\top x$ depends on the input x ; the variance σ^2 does not. For a given x , this is a proper Gaussian density over y .

Example 1.2 (Our residual model). Suppose we have a dynamical system $\dot{x} = f(x)$ and we have partitioned phase space into N clusters. Each cluster k carries a **local model** $L_k: \mathbb{R}^d \rightarrow \mathbb{R}^d$ —a function that, given any point x , outputs a *prediction* of the vector field $f(x)$. The local model is a deterministic function defined by the cluster’s parameters; it is *not* a random variable.

Concretely, L_k is an approximation of f that is built from the cluster’s center c_k and some local dynamical parameters (a Jacobian matrix, an affine operator, or a Koopman operator—the exact form will be specified in Section 6). For now, all that matters is: **given c_k and the local parameters, $L_k(x)$ is a computable function that takes a point $x \in \mathbb{R}^d$ and returns a vector in $\mathbb{R}^d$.** For example, in the simplest case (Taylor linearization):

$$L_k(x) = f(c_k) + \mathbf{J}(c_k)(x - c_k), \quad (5)$$

where $f(c_k) \in \mathbb{R}^d$ is the vector field evaluated at the center (a known vector) and $\mathbf{J}(c_k) \in \mathbb{R}^{d \times d}$ is the Jacobian matrix at the center (a known matrix). The computation is: subtract the center, multiply by the Jacobian, add the vector-field value at the center. The result is an affine function of x .

With L_k defined, we can specify the conditional distribution:

$$f_{\text{obs}}(x_i) | x_i, z_i = k \sim \mathcal{N}(L_k(x_i), \sigma^2 I). \quad (6)$$

This says: given the input x_i and the assignment to cluster k , the observed vector field $f_{\text{obs}}(x_i)$ is a Gaussian random variable with:

- **Center (“mean”)** $= L_k(x_i)$. This is the local model’s prediction of f at x_i . It is computed by evaluating the function L_k at the specific input x_i —**it is a single deterministic value, not an average over anything**. Different inputs give different values.

We call it “the mean” only because of Gaussian convention: in $\mathcal{N}(y; \mu, \Sigma)$, the parameter μ is traditionally called “the mean” because if one drew infinitely many samples $y_1, y_2, \dots$ from this distribution, their sample average would converge to μ . But μ itself is just a location parameter—the point where the bell curve is centered. In our context, a less ambiguous name is **the predicted value**: $L_k(x_i)$ is the value that the local model predicts the vector field should take at x_i , and the Gaussian is centered there.

- **Covariance** $= \sigma^2 I$. This is a fixed isotropic matrix that does *not* depend on x_i . It sets the scale of “tolerable prediction error”—how far the actual observation $f_{\text{obs}}(x_i)$ can be from the prediction $L_k(x_i)$ before the likelihood becomes small.

For a given x_i and k , this is a proper Gaussian density over $f_{\text{obs}} \in \mathbb{R}^d$ (integrates to 1 over $\mathbb{R}^d$).

When we evaluate this density at the *observed* value $f_{\text{obs}}(x_i)$, we get a **likelihood value**:

$$\mathcal{N}(f_{\text{obs}}(x_i); L_k(x_i), \sigma^2 I) = (2\pi\sigma^2)^{-d/2} \exp\left(-\frac{\|f_{\text{obs}}(x_i) - L_k(x_i)\|^2}{2\sigma^2}\right). \quad (7)$$

This number depends on three things:

1. $f_{\text{obs}}(x_i)$: the observed vector field (data—fixed).
2. $L_k(x_i)$: the local model’s prediction (computed from current cluster parameters and input x_i).
3. σ^2 : the noise scale (hyperparameter).

It measures “how well does cluster k ’s local model predict the vector field at x_i ?” A small squared error $\|f_{\text{obs}}(x_i) - L_k(x_i)\|^2$ gives a large likelihood; a large squared error gives a small likelihood.

2 Latent Variable Models and the Marginal Likelihood

2.1 Why latent variables?

A **mixture model** assumes that each data point x_i was generated by one of N components, but we don’t observe which one. The identity of the generating component is a **latent (hidden) variable** $z_i \in \{1, \dots, N\}$.

The generative process for a standard Gaussian mixture:

1. Draw $z_i \sim \text{Categorical}(\pi_1, \dots, \pi_N)$, where $\pi_k \geq 0$, $\sum_k \pi_k = 1$.
2. Draw $x_i \sim \mathcal{N}(\mu_{z_i}, \Sigma_{z_i})$.

We observe x_i but not z_i . To write the density of the observed variable x_i alone, we need to **marginalize out** the latent variable z_i .

Definition 2.1 (Marginalization). If two variables A and B have a joint distribution $p(A, B)$, the **marginal distribution** of A is obtained by summing (or integrating) over all possible values of B :

$$p(A) = \sum_{\text{all } b} p(A, B = b). \quad (8)$$

This “removes” B from the joint by accounting for every possibility.

In our case, $A = x_i$ (observed) and $B = z_i$ (latent, takes values $1, \dots, N$). The marginalization formula gives:

$$p(x_i | \theta) = \sum_{k=1}^N p(x_i, z_i = k | \theta). \quad (9)$$

Each term $p(x_i, z_i = k | \theta)$ is the joint probability of observing x_i *and* having it come from component k . We can decompose this joint using the **chain rule** (also called the product rule) of probability:

$$p(x_i, z_i = k | \theta) = p(z_i = k | \theta) \cdot p(x_i | z_i = k, \theta). \quad (10)$$

This says: the probability of “ x_i was generated by component k ” equals the probability of “component k was chosen” times the probability of “component k generated x_i .” Now we substitute what we know about each factor:

- $p(z_i = k | \theta) = \pi_k$ (by the Categorical prior on z_i).
- $p(x_i | z_i = k, \theta) = \mathcal{N}(x_i; \mu_k, \Sigma_k)$ (by the Gaussian emission model of component k).

Substituting into (10) and then into (9):

$$p(x_i | \theta) = \sum_{k=1}^N \pi_k \mathcal{N}(x_i; \mu_k, \Sigma_k). \quad (11)$$

This is the **marginal density** of x_i —a weighted sum of Gaussians, where $\theta = (\pi, \{\mu_k, \Sigma_k\}_{k=1}^N)$ collects all parameters. Each term contributes the density of x_i under component k , weighted by the prior probability π_k of that component.

2.2 Why not just maximize the marginal likelihood?

The log of the marginal likelihood for all data is

$$\log p(X | \theta) = \sum_{i=1}^P \log \left(\sum_{k=1}^N \pi_k \mathcal{N}(x_i; \mu_k, \Sigma_k) \right). \quad (12)$$

This expression contains a log of a sum, and we claim it has no closed-form maximizer. Let us see concretely why.

For a single Gaussian ($N = 1$), there is no problem. The sum has one term, so

$$\log p(X | \theta) = \sum_i \log \mathcal{N}(x_i; \mu, \Sigma).$$

The log applies directly to a single Gaussian—no sum inside. The derivative with respect to μ is:

$$\frac{\partial}{\partial \mu} \sum_i \log \mathcal{N}(x_i; \mu, \Sigma) = \sum_i \Sigma^{-1}(x_i - \mu) = 0 \implies \mu = \frac{1}{P} \sum_i x_i.$$

One equation, one unknown, closed-form solution.

For a mixture ($N > 1$), the parameters are coupled. Now differentiate with respect to μ_k using the chain rule for $\log(\sum_j \dots)$:

$$\frac{\partial}{\partial \mu_k} \log \left(\sum_j \pi_j \mathcal{N}(x_i; \mu_j, \Sigma_j) \right) = \frac{\pi_k \Sigma_k^{-1}(x_i - \mu_k) \mathcal{N}(x_i; \mu_k, \Sigma_k)}{\sum_j \pi_j \mathcal{N}(x_i; \mu_j, \Sigma_j)}. \quad (13)$$

The denominator $\sum_j \pi_j \mathcal{N}(x_i; \mu_j, \Sigma_j)$ contains **every cluster's parameters**—all μ_j, Σ_j, π_j . Setting the derivative to zero for μ_k gives an equation that involves $\mu_1, \dots, \mu_N, \Sigma_1, \dots, \Sigma_N$, and $\pi_1, \dots, \pi_N$ simultaneously. It is a system of coupled nonlinear equations with no closed-form solution.

Remark 2.1. The ratio $\pi_k \mathcal{N}(x_i; \mu_k, \Sigma_k) / \sum_j \pi_j \mathcal{N}(x_i; \mu_j, \Sigma_j)$ is exactly the responsibility r_{ik} . But here it is a *function of all parameters*, not a fixed number. That is the coupling: you cannot solve for μ_k without knowing r_{ik} , and you cannot compute r_{ik} without knowing all the μ_j . EM breaks this circularity by alternating: fix r_{ik} (E-step), then solve for μ_k (M-step), repeat.

Key idea. Latent variable models have intractable marginal likelihoods because the log of a sum couples all cluster parameters together. The EM algorithm breaks this coupling by replacing $\log(\text{sum})$ with $\text{sum}(\log)$ (via the ELBO), at the cost of introducing an auxiliary distribution q that must be iteratively updated.

3 The Evidence Lower Bound (ELBO)

3.1 The problem: a log of a sum

Our goal is to maximize $\log p(X | \theta)$ (12) over θ . But this expression contains a *logarithm of a sum*, which has no closed-form maximizer. We need a workaround.

The strategy: construct a simpler function—a **lower bound** on $\log p(X | \theta)$ —that *is* tractable to maximize. If we can make this bound tight (close to the true value), then maximizing the bound approximately maximizes the true objective.

3.2 Introducing a helper distribution

The key trick is to introduce an auxiliary distribution q over the latent variables. Let $q(z_i)$ be *any* probability distribution over $\{1, \dots, N\}$ —one for each data point i . We write $q(z_i = k)$ for the probability that q assigns point i to cluster k . The only requirements are $q(z_i = k) \geq 0$ and $\sum_k q(z_i = k) = 1$.

Intuition 3.1. Think of q as our current “guess” at the cluster assignments. It does not have to be the true posterior—it can be *anything*. The ELBO derivation works for any q , and we will later choose q to make the bound as tight as possible.

3.3 Step-by-step derivation

We start from the marginal log-likelihood for a single data point and work through five algebraic steps.

Step 1: rewrite using marginalization. The marginal $p(x_i | \theta)$ is a sum over the latent variable (as derived in Section 2.1):

$$\log p(x_i | \theta) = \log \sum_{k=1}^N p(x_i, z_i = k | \theta). \quad (14)$$

This is just the log of the marginal density. So far, nothing new.

Step 2: multiply and divide by $q(z_i = k)$ inside the sum. For any q with $q(z_i = k) > 0$ for all k :

$$\log \sum_k p(x_i, z_i = k | \theta) = \log \sum_k q(z_i = k) \cdot \frac{p(x_i, z_i = k | \theta)}{q(z_i = k)}. \quad (15)$$

This is a purely algebraic identity—multiplying and dividing by the same positive number changes nothing. **Why do we do this?** Because the sum $\sum_k q(z_i = k) \cdot (\dots)$ is a **weighted average** (an expectation under q), and weighted averages interact nicely with concave functions like log via Jensen’s inequality.

Step 3: apply Jensen’s inequality to get a lower bound. Before stating the inequality, let us understand **why we want a lower bound at all**.

Our goal is to find the parameters θ that maximize $\log p(X | \theta)$. But we cannot maximize $\log p(X | \theta)$ directly because it contains a log-of-a-sum (equation (14)) that has no closed-form solution.

Suppose instead we had a simpler function $\mathcal{L}(q, \theta)$ with two properties:

1. **It is a lower bound:** $\mathcal{L}(q, \theta) \leq \log p(X | \theta)$ for all q, θ .
2. **It can be made tight:** there exists a choice of q such that $\mathcal{L}(q, \theta) = \log p(X | \theta)$ exactly.

If we have such a function, we can use the following two-step strategy to maximize $\log p(X | \theta)$:

- **Tighten the bound** (choose q to close the gap): after this step, $\mathcal{L}(q, \theta) = \log p(X | \theta)$ —the bound touches the true objective.
- **Push the bound up** (choose θ to increase $\mathcal{L}$): after this step, $\mathcal{L}$ increases. Since the bound was tight before this step, the true objective $\log p(X | \theta)$ must also increase (or stay the same).

Alternating these two steps gives a monotonically non-decreasing sequence $\log p(X | \theta^{(0)}) \leq \log p(X | \theta^{(1)}) \leq \dots$. **This is the EM algorithm:** the first step is the E-step, the second is the M-step, and $\mathcal{L}$ is the ELBO.

So: we need a lower bound not for its own sake, but because it gives us a **tractable surrogate** that we can optimize in place of the intractable true objective, with the guarantee that each optimization cycle improves (or maintains) the true objective.

Now we construct such a bound using Jensen’s inequality.

Definition 3.1 (Jensen’s inequality). If φ is a **concave** function (curves downward, like $\log$) and $W_1, \dots, W_N$ are non-negative weights summing to 1, then for any values $a_1, \dots, a_N$:

$$\varphi\left(\sum_k W_k a_k\right) \geq \sum_k W_k \varphi(a_k). \quad (16)$$

In words: the function of the weighted average is $\geq$ the weighted average of the function.

We apply (16) to (15) with $\varphi = \log$ (which is concave), $W_k = q(z_i = k)$, and $a_k = p(x_i, z_i = k | \theta) / q(z_i = k)$:

$$\log \sum_k q(z_i = k) \cdot \frac{p(x_i, z_i = k | \theta)}{q(z_i = k)} \geq \sum_k q(z_i = k) \cdot \log \frac{p(x_i, z_i = k | \theta)}{q(z_i = k)}. \quad (17)$$

The log has moved **inside** the sum, at the cost of turning an equality into an inequality ($\geq$). The right-hand side is a lower bound on the left-hand side.

Step 4: split the log of a ratio. Using $\log(A/B) = \log A - \log B$:

$$\begin{aligned} \sum_k q(z_i = k) \log \frac{p(x_i, z_i = k | \theta)}{q(z_i = k)} &= \sum_k q(z_i = k) \log p(x_i, z_i = k | \theta) \\ &\quad - \sum_k q(z_i = k) \log q(z_i = k). \end{aligned} \quad (18)$$

The first sum is the expected joint log-likelihood under q . The second sum (with the minus sign) is the **entropy** of q , written $H[q] = -\sum_k q_k \log q_k$.

Step 5: sum over all data points and define the ELBO. Summing (17)–(18) over $i = 1, \dots, P$, and writing $r_{ik} = q(z_i = k)$:

$$\log p(X | \theta) \geq \underbrace{\sum_{i,k} r_{ik} \log p(x_i, z_i = k | \theta)}_{\text{expected complete-data log-likelihood}} + \underbrace{\left(-\sum_{i,k} r_{ik} \log r_{ik}\right)}_{\text{entropy } H[q]} \equiv \text{ELBO}(q, \theta). \quad (19)$$

Key idea. The ELBO is a **lower bound** on the log marginal likelihood: $\log p(X | \theta) \geq \text{ELBO}(q, \theta)$ for *any* distribution q over the latent variables.

This bound holds because Jensen’s inequality introduced a $\geq$ at Step 3. The bound is tight (equality holds) when $q(z_i = k) = p(z_i = k | x_i, \theta)$ —i.e., when q equals the true posterior. In that case the gap is zero:

$$\log p(X | \theta) - \text{ELBO}(q, \theta) = \text{KL}(q(Z) \| p(Z | X, \theta)) \geq 0, \quad (20)$$

where KL is the Kullback-Leibler divergence (a non-negative measure of “distance” between two distributions; it equals zero if and only if the two distributions are identical).

Intuition 3.2. Why does this help? Because the ELBO has $\log p(x_i, z_i = k \mid \theta)$ (joint) instead of $\log \sum_k p(x_i, z_i = k \mid \theta)$ (marginal). The joint factors as $p(z_i = k) \cdot p(x_i \mid z_i = k)$ (chain rule), and the log of a product is a sum of logs—no more “log of a sum.” This makes the M-step tractable.

3.4 Decomposing the ELBO

To make the ELBO concrete, we need to expand $\log p(x_i, z_i = k \mid \theta)$. Using the chain rule (10):

$$p(x_i, z_i = k \mid \theta) = p(z_i = k \mid \theta) \cdot p(x_i \mid z_i = k, \theta). \quad (21)$$

Substituting our specific distributions (Categorical prior π_k , Gaussian emission):

$$p(x_i, z_i = k \mid \theta) = \pi_k \cdot \mathcal{N}(x_i; \mu_k, \Sigma_k). \quad (22)$$

Taking the log: $\log p(x_i, z_i = k \mid \theta) = \log \pi_k + \log \mathcal{N}(x_i; \mu_k, \Sigma_k)$. Plugging into the ELBO (19):

$$\text{ELBO}(q, \theta) = \sum_{i,k} r_{ik} \log \pi_k + \sum_{i,k} r_{ik} \log \mathcal{N}(x_i; \mu_k, \Sigma_k) - \sum_{i,k} r_{ik} \log r_{ik}, \quad (23)$$

where we wrote $r_{ik} = q(z_i = k)$. The three terms are:

1. **Assignment log-likelihood:** $\sum_{i,k} r_{ik} \log \pi_k$. Prefers configurations where points are assigned to clusters with large π_k .
2. **Data log-likelihood:** $\sum_{i,k} r_{ik} \log \mathcal{N}(x_i; \mu_k, \Sigma_k)$. Prefers configurations where each point is close to its assigned center.
3. **Entropy:** $-\sum_{i,k} r_{ik} \log r_{ik}$. Prefers “soft” (uncertain) assignments. This prevents the trivial solution of putting all mass on one cluster.

4 The EM Algorithm

4.1 E-step: optimize over q

Fix θ . We want $q^* = \arg \max_q \text{ELBO}(q, \theta)$. Since the gap is the KL divergence $\text{KL}(q \parallel p(Z \mid X, \theta))$, setting $q^*(z_i = k) = p(z_i = k \mid x_i, \theta)$ makes $\text{KL} = 0$ and the ELBO equals $\log p(X \mid \theta)$ exactly.

We compute this using **Bayes’ theorem**.

Definition 4.1 (Bayes’ theorem). For any two events or variables A and B with $p(B) > 0$:

$$p(A \mid B) = \frac{p(B \mid A) p(A)}{p(B)}. \quad (24)$$

In words: posterior = (likelihood $\times$ prior) / evidence. The evidence $p(B)$ is a normalizing constant that ensures the posterior sums/integrates to 1.

We apply (24) with $A = (z_i = k)$ and $B = x_i$:

$$p(z_i = k \mid x_i, \theta) = \frac{p(x_i \mid z_i = k, \theta) \cdot p(z_i = k \mid \theta)}{p(x_i \mid \theta)}. \quad (25)$$

Now we substitute each factor using what we know about our model:

- **Prior:** $p(z_i = k \mid \theta) = \pi_k$ (Categorical distribution over components).

- **Likelihood:** $p(x_i \mid z_i = k, \theta) = \mathcal{N}(x_i; \mu_k, \Sigma_k)$ (Gaussian emission of component k).
- **Evidence:** $p(x_i \mid \theta) = \sum_{j=1}^N p(x_i, z_i = j \mid \theta) = \sum_{j=1}^N \pi_j \mathcal{N}(x_i; \mu_j, \Sigma_j)$ (the marginal density, obtained by summing the joint over all components, as derived in Section 2.1).

Substituting into (25):

$$r_{ik} = p(z_i = k \mid x_i, \theta) = \frac{\pi_k \mathcal{N}(x_i; \mu_k, \Sigma_k)}{\sum_{j=1}^N \pi_j \mathcal{N}(x_i; \mu_j, \Sigma_j)}. \quad (26)$$

Intuition 4.1. The responsibility r_{ik} is the posterior probability that point i belongs to cluster k . It is large when x_i is close to μ_k (high $\mathcal{N}$ value) and π_k is large (component is a priori likely). Points equidistant from two centers will have $r_{ik} \approx r_{ij} \approx 0.5$ —soft assignment.

4.2 M-step: optimize over θ

Fix q (i.e., fix the responsibilities r_{ik}). We want $\theta^* = \arg \max_{\theta} \text{ELBO}(q, \theta)$. Since the entropy term $H[q] = -\sum_{i,k} r_{ik} \log r_{ik}$ does not depend on θ (the r_{ik} are fixed numbers), this is equivalent to maximizing the **expected complete-data log-likelihood**:

$$Q(\theta) = \sum_{i,k} r_{ik} \log p(x_i, z_i = k \mid \theta). \quad (27)$$

Now we expand $\log p(x_i, z_i = k \mid \theta)$ using the chain rule:

$$\begin{aligned} \log p(x_i, z_i = k \mid \theta) &= \log[p(z_i = k \mid \theta) \cdot p(x_i \mid z_i = k, \theta)] \\ &= \log p(z_i = k \mid \theta) + \log p(x_i \mid z_i = k, \theta) \quad (\text{log of a product} = \text{sum of logs}) \\ &= \log \pi_k + \log \mathcal{N}(x_i; \mu_k, \Sigma_k). \quad (\text{substituting our specific distributions}) \end{aligned} \quad (28)$$

Plugging back into (27):

$$Q(\theta) = \sum_{i,k} r_{ik} [\log \pi_k + \log \mathcal{N}(x_i; \mu_k, \Sigma_k)]. \quad (29)$$

Update for π_k : Maximize $\sum_k (\sum_i r_{ik}) \log \pi_k$ subject to $\sum_k \pi_k = 1$. By Lagrange multipliers:

$$\pi_k^{\text{new}} = \frac{R_k}{P}, \quad R_k = \sum_{i=1}^P r_{ik}. \quad (30)$$

R_k is the **effective mass** of cluster k —the total soft-assignment weight. The updated π_k is just the fraction of total mass in cluster k .

Update for μ_k : Maximize $\sum_i r_{ik} \log \mathcal{N}(x_i; \mu_k, \Sigma_k)$ over μ_k . The log-Gaussian contains $-\frac{1}{2}(x_i - \mu_k)^\top \Sigma_k^{-1} (x_i - \mu_k)$. Taking the gradient:

$$\nabla_{\mu_k} Q = \sum_i r_{ik} \Sigma_k^{-1} (x_i - \mu_k) = 0 \implies \mu_k^{\text{new}} = \frac{\sum_i r_{ik} x_i}{R_k}. \quad (31)$$

The new center is the **responsibility-weighted mean** of the data.

Update for Σ_k : Maximize over Σ_k . This involves the log-determinant and the quadratic form. Taking the matrix derivative and setting to zero:

$$\Sigma_k^{\text{new}} = \frac{1}{R_k} \sum_i r_{ik} (x_i - \mu_k^{\text{new}})(x_i - \mu_k^{\text{new}})^\top. \quad (32)$$

The new covariance is the **responsibility-weighted scatter** around the new mean.

Key idea. EM monotonicity: Each E-step makes $\text{ELBO}(q, \theta) = \log p(X | \theta)$ (closes the KL gap). Each M-step maximizes $Q(\theta)$ over θ , which increases ELBO (and hence $\log p(X | \theta)$). The sequence of marginal likelihoods is non-decreasing:

$$\log p(X | \theta^{(0)}) \leq \log p(X | \theta^{(1)}) \leq \log p(X | \theta^{(2)}) \leq \dots$$

Convergence to a local maximum is guaranteed.

5 Adding Bayesian Priors

5.1 Why priors?

MLE can overfit: with N clusters and P points, if a cluster contains very few points, its covariance estimate $\hat{\Sigma}_k$ becomes singular (not invertible). Bayesian priors **regularize** by encoding prior beliefs about parameter ranges. The ELBO becomes:

$$\text{ELBO}(q, \theta) = \mathbb{E}_q[\log p(X, Z | \theta)] + H[q] + \log p(\theta), \quad (33)$$

where $\log p(\theta)$ is the log-prior on all parameters. The M-step now maximizes $Q(\theta) + \log p(\theta)$, which is the **MAP (maximum a posteriori)** objective.

5.2 The Dirichlet prior on mixing weights

The mixing weights $\pi = (\pi_1, \dots, \pi_N)$ live on the probability simplex ($\pi_k \geq 0$, $\sum_k \pi_k = 1$). The natural prior is the **Dirichlet distribution**:

$$p(\pi | \alpha_0) = \frac{\Gamma(N\alpha_0)}{\Gamma(\alpha_0)^N} \prod_{k=1}^N \pi_k^{\alpha_0-1}. \quad (34)$$

The parameter $\alpha_0 > 0$ controls the prior's character:

- $\alpha_0 = 1$: uniform prior on the simplex (all weight allocations equally likely).
- $\alpha_0 > 1$: prefers π near uniform ($\pi_k \approx 1/N$ for all k). Discourages empty clusters.
- $\alpha_0 < 1$: **sparse-inducing**. Prefers π concentrated on a few components, with many $\pi_k \approx 0$.

Key idea. We use $\alpha_0 = 0.5 < 1$ to encourage the model to **prune unused clusters**. Starting with N larger than needed, the Dirichlet prior drives $\pi_k \rightarrow 0$ for redundant components. This is automatic model-complexity selection.

MAP update for π : Adding $\log p(\pi) = \sum_k (\alpha_0 - 1) \log \pi_k + \text{const}$ to the Lagrangian:

$$\pi_k^{\text{new}} = \frac{R_k + \alpha_0 - 1}{P + N(\alpha_0 - 1)}. \quad (35)$$

When $\alpha_0 < 1$ and R_k is small, π_k can go negative; we clamp to $\pi_k \geq 10^{-10}$ and renormalize. In practice, clusters with $R_k < 1$ are pruned (removed from the model entirely).

5.3 The Normal-Inverse-Wishart (NIW) prior on centers and covariances

For the center c_k and covariance Σ_k of each cluster, we use the **NIW prior**:

$$\Sigma_k \sim \mathcal{IW}(\Psi_0, \nu_0), \quad (36)$$

$$c_k \mid \Sigma_k \sim \mathcal{N}(\mu_0, \Sigma_k / \kappa_0), \quad (37)$$

where $\Psi_0 \in \mathbb{R}^{d \times d}$ (positive definite scale matrix), $\nu_0 > d - 1$ (degrees of freedom), $\mu_0 \in \mathbb{R}^d$ (prior mean), $\kappa_0 > 0$ (prior precision scaling).

Intuition 5.1. The NIW prior encodes:

- μ_0 : “we expect cluster centers near this point” (set to the data mean).
- κ_0 : “how strongly we believe that” (small κ_0 = weak prior = centers free to move; large κ_0 = strong pull toward μ_0).
- Ψ_0 : “we expect covariances of roughly this scale” (set proportional to identity).
- ν_0 : “how confident we are about the covariance” (small ν_0 = vague prior; large = confident).

Why “conjugate”? A prior is **conjugate** to a likelihood if the posterior has the same functional form as the prior. For a Gaussian likelihood with NIW prior, the posterior on (c_k, Σ_k) is also NIW with updated parameters:

$$\kappa_n = \kappa_0 + R_k, \quad \nu_n = \nu_0 + R_k, \quad (38)$$

$$\mu_n = \frac{\kappa_0 \mu_0 + R_k \bar{x}_k}{\kappa_n}, \quad (39)$$

$$\Psi_n = \Psi_0 + S_k + \frac{\kappa_0 R_k}{\kappa_n} (\bar{x}_k - \mu_0)(\bar{x}_k - \mu_0)^\top, \quad (40)$$

where $\bar{x}_k = \frac{1}{R_k} \sum_i r_{ik} x_i$ is the responsibility-weighted mean and $S_k = \sum_i r_{ik} (x_i - \bar{x}_k)(x_i - \bar{x}_k)^\top$ is the weighted scatter.

Conjugacy is important because it gives **closed-form marginal likelihoods** (Section 9).

MAP updates for c_k and Σ_k : Setting the gradient of $Q(\theta) + \log p(\theta)$ to zero for c_k :

$$(\kappa_0 \Sigma_k^{-1} + R_k \Sigma_k^{-1}) c_k^{\text{new}} = \kappa_0 \Sigma_k^{-1} \mu_0 + \Sigma_k^{-1} \sum_i r_{ik} x_i. \quad (41)$$

Since Σ_k^{-1} factors out (and we often write $\Lambda_0 = \kappa_0 \Sigma_k^{-1}$ or a fixed matrix):

$$c_k^{\text{new}} = \frac{\kappa_0 \mu_0 + \sum_i r_{ik} x_i}{\kappa_0 + R_k} = \frac{\kappa_0 \mu_0 + R_k \bar{x}_k}{\kappa_n}. \quad (42)$$

This is a **weighted average** of the prior mean μ_0 and the data mean $\bar{x}_k$, with weights κ_0 (prior) and R_k (data). More data $\rightarrow$ center follows data; less data $\rightarrow$ center stays near μ_0 .

For Σ_k , the MAP mode of the Inverse-Wishart posterior is:

$$\Sigma_k^{\text{new}} = \frac{\Psi_n}{\nu_n + d + 1} = \frac{\Psi_0 + S_k + \text{conflict}}{\nu_0 + R_k + d + 1}. \quad (43)$$

The “conflict” term $\frac{\kappa_0 R_k}{\kappa_n} (\bar{x}_k - \mu_0)(\bar{x}_k - \mu_0)^\top$ penalizes disagreement between the prior mean and data mean. In our implementation, we use a simplified version where Λ_0 is a fixed precision matrix:

$$\Sigma_k^{\text{new}} = \frac{\Psi_0 + \sum_i r_{ik} (x_i - c_k^{\text{new}})(x_i - c_k^{\text{new}})^\top}{\nu_0 + R_k + d + 1} + \lambda I. \quad (44)$$

The λI term (typically 10^{-6}) prevents numerical singularity.

6 Our Model: Adding the Residual Factor

6.1 The generative model

We augment the standard GMM with a **second observation per data point**: the vector-field value $f(x_i)$. The generative story for each point:

$$z_i \sim \text{Cat}(\pi), \quad (45)$$

$$x_i \mid z_i = k \sim \mathcal{N}(c_k, \Sigma_k), \quad (46)$$

$$f_{\text{obs}}(x_i) \mid x_i, z_i = k \sim \mathcal{N}(L_k(x_i), \sigma^2 I). \quad (47)$$

Line (46) is the proximity model (standard GMM). Line (47) is the residual model (novel): it says that the observed vector field $f_{\text{obs}}(x_i)$ is drawn from a Gaussian centered at the local model’s prediction $L_k(x_i)$, with isotropic noise $\sigma^2 I$.

What is $L_k(x_i)$? This is the local model’s prediction of the vector field at x_i . It depends on:

- The cluster’s center c_k and local-model parameters (e.g., Jacobian $\mathbf{J}_k$, or Koopman operator $\mathbf{M}_k$).
- The input point x_i itself.

Its specific form depends on the variant (Taylor, LS, or EDMD). The framework is agnostic to this choice.

6.2 The joint observation likelihood

For a single data point $(x_i, f_{\text{obs}}(x_i))$ assigned to cluster k , we want the joint density $p(x_i, f_{\text{obs}}(x_i) \mid z_i = k, \theta)$. We build it in three steps.

Step 1: apply the chain rule of probability. For any two variables A and B :

$$p(A, B) = p(A) \cdot p(B \mid A). \quad (48)$$

This decomposes a joint density into a marginal times a conditional. Applying this with $A = x_i$ and $B = f_{\text{obs}}(x_i)$, all conditioned on $z_i = k$:

$$p(x_i, f_{\text{obs}}(x_i) \mid z_i = k, \theta) = \underbrace{p(x_i \mid z_i = k, \theta)}_{\text{factor 1}} \cdot \underbrace{p(f_{\text{obs}}(x_i) \mid x_i, z_i = k, \theta)}_{\text{factor 2}}. \quad (49)$$

Step 2: identify each factor from our generative model.

- **Factor 1** (proximity): From (46), the generative model says $x_i \mid z_i = k \sim \mathcal{N}(c_k, \Sigma_k)$. So:

$$p(x_i \mid z_i = k, \theta) = \mathcal{N}(x_i; c_k, \Sigma_k). \quad (50)$$

- **Factor 2** (residual): From (47), the generative model says $f_{\text{obs}}(x_i) \mid x_i, z_i = k \sim \mathcal{N}(L_k(x_i), \sigma^2 I)$. So:

$$p(f_{\text{obs}}(x_i) \mid x_i, z_i = k, \theta) = \mathcal{N}(f_{\text{obs}}(x_i); L_k(x_i), \sigma^2 I). \quad (51)$$

Step 3: substitute into the chain rule (49).

$$p(x_i, f_{\text{obs}}(x_i) \mid z_i = k, \theta) = \underbrace{\mathcal{N}(x_i; c_k, \Sigma_k)}_{\text{proximity}} \cdot \underbrace{\mathcal{N}(f_{\text{obs}}(x_i); L_k(x_i), \sigma^2 I)}_{\text{residual}}. \quad (52)$$

This is a product of two Gaussian densities over **different random variables**: the first is a density over $x_i \in \mathbb{R}^d$, the second is a density over $f_{\text{obs}}(x_i) \in \mathbb{R}^d$. Together they define a density over the pair $(x_i, f_{\text{obs}}(x_i)) \in \mathbb{R}^{2d}$.

Derivation. Why is (48) valid here? In the generative model, x_i is drawn first (from the proximity model), then $f_{\text{obs}}(x_i)$ is drawn conditionally on x_i (from the residual model). The chain rule of probability is a universal identity—it holds for any joint distribution, not just ours. We are simply choosing to decompose $p(x_i, f_{\text{obs}})$ into $p(x_i) \cdot p(f_{\text{obs}} \mid x_i)$ rather than the other way around, because this matches the order of our generative process.

6.3 The log-likelihood of the joint observation

Taking the log:

$$\begin{aligned} \log p(x_i, f_{\text{obs}}(x_i) \mid z_i = k) &= \underbrace{-\frac{d}{2} \log 2\pi - \frac{1}{2} \log |\Sigma_k| - \frac{1}{2} (x_i - c_k)^\top \Sigma_k^{-1} (x_i - c_k)}_{\text{proximity: depends on } x_i, c_k, \Sigma_k} \\ &\quad + \underbrace{-\frac{d}{2} \log(2\pi\sigma^2) - \frac{\|f_{\text{obs}}(x_i) - L_k(x_i)\|^2}{2\sigma^2}}_{\text{residual: depends on } f_{\text{obs}}(x_i), L_k(x_i), \sigma^2}. \end{aligned} \quad (53)$$

Key idea. The log-likelihood is the **sum of two quadratic penalties**:

1. **Proximity penalty:** $(x_i - c_k)^\top \Sigma_k^{-1} (x_i - c_k)$ — how far is x_i from the center in the Σ_k -metric?
2. **Residual penalty:** $\|f_{\text{obs}}(x_i) - L_k(x_i)\|^2 / \sigma^2$ — how badly does the local model predict f at this point?

The relative importance of these two penalties is controlled by the ratio of their scales: $\lambda(\Sigma_k^{-1})$ vs. $1/\sigma^2$. When σ^2 is small, the residual penalty dominates; when σ^2 is large, the proximity penalty dominates.

6.4 The role of the residual scale

σ^2 is the single most important hyperparameter in the framework. It controls the **balance between the two forces**:

- $\sigma^2 \rightarrow \infty$: The residual penalty $\|\varepsilon_k\|^2/(2\sigma^2) \rightarrow 0$. The joint likelihood reduces to the proximity factor alone. **You recover standard GMM.** Clusters form purely by geometry, ignoring dynamics.
- $\sigma^2 \rightarrow 0$: The residual penalty dominates everything. A point is assigned to whichever cluster has the smallest $\|\varepsilon_k\|$, regardless of geometric proximity. **You get a piecewise regression** where clusters form by model fit alone.
- $\sigma^2 \sim \text{median}(\|\varepsilon_k\|^2)/d$: Both forces contribute comparably. **This is the interesting regime.** Clusters form by a combination of geometric proximity and model quality.

Auto-calibration: We set σ^2 from the initial GMM partition’s residuals:

$$\sigma^2 = \max\left(\frac{\text{median}_i \|\varepsilon_k(x_i)\|^2}{d}, 10^{-3}\right). \quad (54)$$

This ensures the residual penalty is on the same scale as the actual linearization errors in the initial (geometric) partition. The floor 10^{-3} prevents numerical instability when residuals are very small (e.g., on polynomial systems where the local model is exact).

Could σ^2 be learned? Yes. The M-step update would be:

$$\sigma^{2,\text{new}} = \frac{\sum_{i,k} r_{ik} \|\varepsilon_k(x_i)\|^2}{d \cdot P}. \quad (55)$$

But this can collapse to $\sigma^2 \rightarrow \infty$ (residual becomes irrelevant) or $\sigma^2 \rightarrow 0$ (proximity becomes irrelevant), both of which are local optima of the ELBO. A prior $\sigma^2 \sim \text{InverseGamma}(a_0, b_0)$ would prevent collapse but adds two more hyperparameters. We defer this and use the auto-calibrated fixed value.

7 E-step for Our Model

The E-step computes the posterior $p(z_i = k \mid x_i, f_{\text{obs}}(x_i), \theta)$ —the probability that point i belongs to cluster k , given *both* the position x_i and the observed vector field $f_{\text{obs}}(x_i)$. We derive this in three steps.

Step 1: apply Bayes’ theorem (24). Set $A = (z_i = k)$ and $B = (x_i, f_{\text{obs}}(x_i))$ (i.e., the “data” is now the *pair* of observations):

$$p(z_i = k \mid x_i, f_{\text{obs}}(x_i), \theta) = \frac{p(x_i, f_{\text{obs}}(x_i) \mid z_i = k, \theta) \cdot p(z_i = k \mid \theta)}{p(x_i, f_{\text{obs}}(x_i) \mid \theta)}. \quad (56)$$

Step 2: identify each factor from our model.

- **Likelihood:** $p(x_i, f_{\text{obs}}(x_i) \mid z_i = k, \theta)$ is the joint observation likelihood (52), which we already derived: $= \mathcal{N}(x_i; c_k, \Sigma_k) \cdot \mathcal{N}(f_{\text{obs}}(x_i); L_k(x_i), \sigma^2 I)$.
- **Prior:** $p(z_i = k \mid \theta) = \pi_k$ (Categorical).
- **Evidence:** $p(x_i, f_{\text{obs}}(x_i) \mid \theta) = \sum_{j=1}^N p(x_i, f_{\text{obs}}(x_i), z_i = j \mid \theta)$ (marginalize over all clusters, same pattern as Section 2.1). Using the chain rule on each term: $= \sum_{j=1}^N \pi_j \cdot \mathcal{N}(x_i; c_j, \Sigma_j) \cdot \mathcal{N}(f_{\text{obs}}(x_i); L_j(x_i), \sigma^2 I)$.

Step 3: substitute into (56). Writing $\varepsilon_k(x_i) = f_{\text{obs}}(x_i) - L_k(x_i)$:

$$r_{ik} = p(z_i = k \mid x_i, f_{\text{obs}}(x_i), \theta) = \frac{\pi_k \mathcal{N}(x_i; c_k, \Sigma_k) \mathcal{N}(\varepsilon_k(x_i); 0, \sigma^2 I)}{\sum_{j=1}^N \pi_j \mathcal{N}(x_i; c_j, \Sigma_j) \mathcal{N}(\varepsilon_j(x_i); 0, \sigma^2 I)}. \quad (57)$$

Compare to standard GMM (26): the only difference is the extra factor $\mathcal{N}(\varepsilon_k(x_i); 0, \sigma^2 I)$ in both numerator and denominator. This factor **upweights** clusters where the residual is small (good local fit) and **downweights** clusters where the residual is large (bad local fit).

Numerical implementation: We work in log-space to avoid underflow:

$$\log \tilde{r}_{ik} = \log \pi_k + \log \mathcal{N}(x_i; c_k, \Sigma_k) + \log \mathcal{N}(\varepsilon_k(x_i); 0, \sigma^2 I), \quad (58)$$

$$\log r_{ik} = \log \tilde{r}_{ik} - \underset{k}{\text{logsumexp}} \log \tilde{r}_{ik}, \quad (59)$$

where $\text{logsumexp}_k(a_k) = \log \sum_k \exp(a_k)$, computed via the max-shift trick for numerical stability.

8 M-step for Our Model

The M-step maximizes the expected complete-data log-likelihood $Q(\theta)$ plus log-priors over θ . We build Q step by step.

The general form of Q is (27): $Q(\theta) = \sum_{i,k} r_{ik} \log p(x_i, f_{\text{obs}}(x_i), z_i = k \mid \theta)$. Using the chain rule to decompose the joint:

$$\begin{aligned} \log p(x_i, f_{\text{obs}}(x_i), z_i = k \mid \theta) &= \log p(z_i = k \mid \theta) + \log p(x_i \mid z_i = k, \theta) + \log p(f_{\text{obs}}(x_i) \mid x_i, z_i = k, \theta) \\ &= \log \pi_k + \log \mathcal{N}(x_i; c_k, \Sigma_k) + \log \mathcal{N}(\varepsilon_k(x_i); 0, \sigma^2 I). \end{aligned} \quad (60)$$

The first two terms are identical to standard GMM. The third term—the log-residual—is our novel contribution. Substituting into Q :

$$Q(\theta) = \underbrace{\sum_{i,k} r_{ik} \log \pi_k}_{Q_\pi} + \underbrace{\sum_{i,k} r_{ik} \log \mathcal{N}(x_i; c_k, \Sigma_k)}_{Q_{\text{prox}}} + \underbrace{\sum_{i,k} r_{ik} \log \mathcal{N}(\varepsilon_k(x_i); 0, \sigma^2 I)}_{Q_{\text{res}}}. \quad (61)$$

The Q_π and Q_{prox} terms are identical to standard GMM. The novel Q_{res} term is:

$$Q_{\text{res}} = -\frac{Pd}{2} \log(2\pi\sigma^2) - \frac{1}{2\sigma^2} \sum_{i,k} r_{ik} \|\varepsilon_k(x_i)\|^2. \quad (62)$$

The π and Σ_k updates are the same as in Sections 4–5. What changes is the **center update**: Q_{res} depends on c_k through $\varepsilon_k(x_i) = f(x_i) - L_k(x_i)$, and L_k depends on c_k . How this dependence is handled defines the three variants.

8.1 Variant A: Taylor-analytic

Here $L_k(x) = f(c_k) + \mathbf{J}(c_k)(x - c_k)$, where f and $\mathbf{J}$ are evaluated analytically. The center c_k enters Q_{res} through both $f(c_k)$ and $\mathbf{J}(c_k)$. In the M-step, we:

1. Update c_k using only the proximity gradient:

$$(\Lambda_0 + R_k \Sigma_k^{-1}) c_k^{\text{new}} = \Lambda_0 \mu_0 + \Sigma_k^{-1} \sum_i r_{ik} x_i. \quad (63)$$

This is the standard GMM center update with NIW prior.

2. **Re-tie**: set $f(c_k) \leftarrow f(c_k^{\text{new}})$, $\mathbf{J}(c_k) \leftarrow \mathbf{J}(c_k^{\text{new}})$ analytically.

Remark 8.1 (Why this is approximate). The gradient $\nabla_{c_k} Q_{\text{res}}$ is nonzero—it contains a term from $\partial \varepsilon_k / \partial c_k$. By ignoring it in step 1, we are not fully maximizing Q over c_k . This makes the update a **generalized EM (GEM)** step: it increases Q but does not maximize it. The ELBO is still non-decreasing at each step (GEM guarantees this), but it may not converge to the same point as full maximization.

Additionally, step 2 “re-ties” $f(c_k)$ and $\mathbf{J}(c_k)$ to the analytic values at the new center. This changes the residual in a way that is *not* accounted for by the gradient—it’s a separate heuristic update. In practice, at small N (large clusters, large σ^2), this is harmless. At large N (small clusters, small σ^2), the residual term dominates and the non-variational re-tying can decrease the ELBO.

8.2 Variant B: Taylor-LS

Here $L_k(x) = f_k + \mathbf{J}_k(x - c_k)$, where $(f_k, \mathbf{J}_k)$ are **free parameters** fit by weighted least squares. We now have three parameter groups to update: $(f_k, \mathbf{J}_k)$, c_k , and Σ_k . We update them sequentially (coordinate ascent):

Step 1: update $(f_k, \mathbf{J}_k)$ by weighted LS. The residual is $\varepsilon_k(x_i) = f_{\text{obs}}(x_i) - f_k - \mathbf{J}_k(x_i - c_k)$. Fix c_k . The Q_{res} term is:

$$Q_{\text{res}}(f_k, \mathbf{J}_k) = -\frac{1}{2\sigma^2} \sum_i r_{ik} \|f_{\text{obs}}(x_i) - f_k - \mathbf{J}_k(x_i - c_k)\|^2 + \text{const}. \quad (64)$$

This is a **weighted least-squares** problem. Define:

$$\phi_i = \begin{bmatrix} 1 \\ x_i - c_k \end{bmatrix} \in \mathbb{R}^{d+1}, \quad Z = \begin{bmatrix} \phi_1^\top \\ \vdots \\ \phi_P^\top \end{bmatrix} \in \mathbb{R}^{P \times (d+1)}, \quad (65)$$

$$W = \text{diag}(r_{1k}, \dots, r_{Pk}), \quad Y = \begin{bmatrix} f_{\text{obs}}(x_1)^\top \\ \vdots \\ f_{\text{obs}}(x_P)^\top \end{bmatrix} \in \mathbb{R}^{P \times d}. \quad (66)$$

The optimal $B = [f_k^\top; \mathbf{J}_k^\top] \in \mathbb{R}^{(d+1) \times d}$ satisfies the normal equations:

$$(Z^\top W Z + \lambda I) B = Z^\top W Y. \quad (67)$$

The ridge λI (typically $\lambda = 10^{-6}$) regularizes against ill-conditioning when R_k is small.

Derivation. Take $\nabla_{f_k} Q_{\text{res}} = 0$:

$$\sum_i r_{ik}(f_{\text{obs}}(x_i) - f_k - \mathbf{J}_k(x_i - c_k)) = 0 \implies R_k f_k = \sum_i r_{ik} f_{\text{obs}}(x_i) - \mathbf{J}_k \sum_i r_{ik}(x_i - c_k).$$

This, together with the analogous equation for $\mathbf{J}_k$, is equivalent to the normal equations (67).

Step 2: update c_k (full gradient, including residual). Now fix the new $(f_k, \mathbf{J}_k)$ and optimize c_k . The Q -function depends on c_k through both the proximity term and the residual term. We need:

$$\nabla_{c_k} Q = \nabla_{c_k} Q_{\text{prox}} + \nabla_{c_k} Q_{\text{res}} + \nabla_{c_k} \log p(c_k) = 0. \quad (68)$$

Derivation. Proximity gradient: Q_{prox} contains $-\frac{1}{2} \sum_i r_{ik}(x_i - c_k)^\top \Sigma_k^{-1}(x_i - c_k)$. Taking the gradient:

$$\nabla_{c_k} Q_{\text{prox}} = \Sigma_k^{-1} \sum_i r_{ik}(x_i - c_k) = \Sigma_k^{-1}(S_x - R_k c_k),$$

where $S_x = \sum_i r_{ik} x_i$.

Residual gradient: Q_{res} contains $-\frac{1}{2\sigma^2} \sum_i r_{ik} \|\varepsilon_k(x_i)\|^2$ with $\varepsilon_k(x_i) = f(x_i) - f_k - \mathbf{J}_k(x_i - c_k)$. Note $\frac{\partial \varepsilon_k}{\partial c_k} = \mathbf{J}_k$ (since $\frac{\partial}{\partial c_k}[-\mathbf{J}_k(x_i - c_k)] = \mathbf{J}_k$). So:

$$\begin{aligned} \nabla_{c_k} Q_{\text{res}} &= -\frac{1}{\sigma^2} \sum_i r_{ik} \mathbf{J}_k^\top \varepsilon_k(x_i) \\ &= -\frac{1}{\sigma^2} \mathbf{J}_k^\top [S_f - R_k f_k - \mathbf{J}_k(S_x - R_k c_k)], \end{aligned} \quad (69)$$

where $S_f = \sum_i r_{ik} f_{\text{obs}}(x_i)$.

Prior gradient: $\log p(c_k) \supset -\frac{1}{2}(c_k - \mu_0)^\top \Lambda_0(c_k - \mu_0)$, so $\nabla_{c_k} \log p(c_k) = \Lambda_0(\mu_0 - c_k)$.

Setting the total gradient to zero:

$$\Lambda_0(\mu_0 - c_k) + \Sigma_k^{-1}(S_x - R_k c_k) - \frac{1}{\sigma^2} \mathbf{J}_k^\top [S_f - R_k f_k - \mathbf{J}_k(S_x - R_k c_k)] = 0. \quad (70)$$

Collecting all terms proportional to c_k :

$$\underbrace{\left[\Lambda_0 + R_k \Sigma_k^{-1} + \frac{R_k}{\sigma^2} \mathbf{J}_k^\top \mathbf{J}_k \right]}_{\text{LHS}} c_k^{\text{new}} = \underbrace{\Lambda_0 \mu_0 + \Sigma_k^{-1} S_x - \frac{1}{\sigma^2} \mathbf{J}_k^\top (S_f - R_k f_k - \mathbf{J}_k S_x)}_{\text{RHS}}. \quad (71)$$

Key idea. Equation (71) has three contributions to the left-hand side (the “effective precision” of the center):

1. Λ_0 : pull toward the prior mean μ_0 .
2. $R_k \Sigma_k^{-1}$: pull toward the data mean S_x/R_k (standard GMM term).
3. $(R_k/\sigma^2) \mathbf{J}_k^\top \mathbf{J}_k$: pull toward the point that minimizes the residual (novel term).

Variant A (Taylor-analytic) drops the third term entirely. This is what makes it approximate.

Step 3: update Σ_k . Same as standard GMM with NIW posterior (no residual dependence on Σ_k):

$$\Sigma_k^{\text{new}} = \frac{\Psi_0 + \sum_i r_{ik}(x_i - c_k^{\text{new}})(x_i - c_k^{\text{new}})^\top}{\nu_0 + R_k + d + 1} + \lambda I. \quad (72)$$

8.3 Variant C: Local EDMD

Here $L_k(x) = [\mathbf{M}_k \Phi(x - c_k)]_{1:d}$, where $\Phi: \mathbb{R}^d \rightarrow \mathbb{R}^M$ is a polynomial lifting and $\mathbf{M}_k \in \mathbb{R}^{M \times M}$ is the local Koopman generator.

What is the polynomial lifting? For degree p in d dimensions, $\Phi(u)$ is the vector of all monomials $u_1^{a_1} \cdots u_d^{a_d}$ with $\sum_j a_j \leq p$. For example, with $d = 2, p = 2$:

$$\Phi(u_1, u_2) = (1, u_1, u_2, u_1^2, u_1 u_2, u_2^2)^\top \in \mathbb{R}^6. \quad (73)$$

The total number of monomials is $M = \binom{d+p}{p}$.

Why $[\mathbf{M}_k \Phi]_{1:d}$ gives $\hat{f}$: The first entry of Φ is the constant 1; entries 2 through $d + 1$ are the linear monomials $u_1, \dots, u_d$. Their time derivatives are $\dot{u}_j = \dot{x}_j = f_j(x)$. The Koopman generator $\mathbf{M}_k$ is fit so that $\mathbf{M}_k \Phi(u) \approx \dot{\Phi}(u)$, where $\dot{\Phi}$ is the vector of time derivatives of all monomials. Entries 2 through $d + 1$ of $\dot{\Phi}$ are exactly $f_1(x), \dots, f_d(x)$. So:

$$[\mathbf{M}_k \Phi(x - c_k)]_j \approx \dot{\Phi}_j = f_j(x), \quad j = 1, \dots, d. \quad (74)$$

M-step for $\mathbf{M}_k$: The generator is fit by weighted least squares. The target for each data point is $\dot{\Phi}(x_i - c_k) = \nabla \Phi(x_i - c_k) \cdot f(x_i)$, the Lie derivative of Φ along f . The feature is $\Phi(x_i - c_k)$. With weights r_{ik} :

$$G_k = \sum_i r_{ik} \Phi(x_i - c_k) \Phi(x_i - c_k)^\top \in \mathbb{R}^{M \times M}, \quad (75)$$

$$A_k = \sum_i r_{ik} \dot{\Phi}(x_i - c_k) \Phi(x_i - c_k)^\top \in \mathbb{R}^{M \times M}, \quad (76)$$

$$\mathbf{M}_k = A_k (G_k + \lambda I)^{-1}. \quad (77)$$

Derivation. The objective is $\min_{\mathbf{M}_k} \sum_i r_{ik} \|\dot{\Phi}_i - \mathbf{M}_k \Phi_i\|^2$. This is a weighted linear regression in $\mathbb{R}^M$: the matrix $\mathbf{M}_k$ is the “weight matrix.” Taking the matrix derivative:

$$\nabla_{\mathbf{M}_k} = -2 \sum_i r_{ik} (\dot{\Phi}_i - \mathbf{M}_k \Phi_i) \Phi_i^\top = 0 \implies \mathbf{M}_k \sum_i r_{ik} \Phi_i \Phi_i^\top = \sum_i r_{ik} \dot{\Phi}_i \Phi_i^\top.$$

This gives $\mathbf{M}_k G_k = A_k$, hence $\mathbf{M}_k = A_k G_k^{-1}$.

c_k update: In this variant we use the proximity-only update (63). The residual gradient w.r.t. c_k involves $\nabla_{c_k} \Phi(x_i - c_k)$, which is a tensor of polynomial derivatives. Deriving the closed-form is possible but algebraically heavy and we defer it. This makes the local-EDMD M-step a GEM step for c_k , analogous to Variant A.

9 Model Selection

9.1 How many clusters?

Given the model with N clusters, we want to compare models with different N . The Bayesian approach computes the **marginal likelihood** $p(X, F \mid N)$ —the probability of the data under model N after integrating out all parameters.

9.2 The NIW marginal likelihood (derivation)

For a single cluster k with points $\{x_i : z_i = k\}$, the marginal likelihood integrates out (c_k, Σ_k) :

$$p(X_k) = \int p(X_k \mid c_k, \Sigma_k) p(c_k, \Sigma_k) dc_k d\Sigma_k. \quad (78)$$

Because the NIW prior is conjugate to the Gaussian likelihood, this integral has a closed form:

$$\begin{aligned} \log p(X_k) = & \frac{d}{2} \log \frac{\kappa_0}{\kappa_n} + \frac{\nu_0}{2} \log |\Psi_0| - \frac{\nu_n}{2} \log |\Psi_n| \\ & + \log \Gamma_d\left(\frac{\nu_n}{2}\right) - \log \Gamma_d\left(\frac{\nu_0}{2}\right) - \frac{R_k d}{2} \log \pi, \end{aligned} \quad (79)$$

where Γ_d is the multivariate gamma function $\Gamma_d(a) = \pi^{d(d-1)/4} \prod_{j=1}^d \Gamma(a + (1-j)/2)$.

In our model, we add the residual penalty term:

$$\log p(X_k, F_k) = \log p(X_k) - \frac{1}{2\sigma^2} \sum_{i: z_i = k} \|\varepsilon_k(x_i)\|^2. \quad (80)$$

The total log marginal is:

$$\log p(X, F \mid N) = \underbrace{\log \frac{\Gamma(N\alpha_0)}{\Gamma(P + N\alpha_0)} \prod_k \frac{\Gamma(R_k + \alpha_0)}{\Gamma(\alpha_0)}}_{\text{Dirichlet-Categorical marginal}} + \sum_k \log p(X_k, F_k). \quad (81)$$

The model with the highest $\log p(X, F \mid N)$ is preferred.

10 Summary: What Each Piece Does

Object	What it controls	How it's determined
π_k	Prior probability of cluster k	MAP with Dirichlet prior (Eq. 35)
c_k	Center of cluster k	MAP with NIW prior + residual gradient (Eq. 71)
Σ_k	Shape/size of cluster k	NIW posterior mode (scatter around c_k)
σ^2	Balance: proximity vs. residual	Auto-calibrated from initial residuals; fixed during EM
L_k (local model)	Vector-field prediction in cluster k	Depends on variant (analytic $\mathbf{J}$, LS fit, or EDMD)
r_{ik}	Soft assignment of point i to cluster k	E-step posterior (Eq. 57)
N	Number of clusters	Set high initially; pruned by Dirichlet prior; selected by marginal likelihood

Table 1: Summary of all quantities in the framework.

Hyperparameter	Meaning	Our choice
α_0	Dirichlet concentration	0.5 (sparse-inducing)
μ_0	Prior mean for centers	Data mean $\bar{x}$
Λ_0 (or κ_0)	Prior precision for centers	$0.01I$ (weak: centers free to move)
Ψ_0	Prior scale for covariances	$10I$ or $1I$ (moderate)
ν_0	Prior df for covariances	$d + 2$ (minimal informative)
σ^2	Residual noise scale	Auto-calibrated from median residual
λ (ridge)	Regularization for LS/EDMD	10^{-6} (Taylor-LS) or 10^{-4} (EDMD)

Table 2: Hyperparameters and default values.